2025 Spring Math

HTOP,PDE,COMPLEX,ALGO

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Contents

Chapter 1

Functions of Complex Variables

§1.1 Lecture 1 (02-03)Complex numbers

§1.1.1 Introduction

You know the sets

$$(R,+,\times) and (Q,+,\times)$$

$Q(\sqrt{2}) = \{a + b\sqrt{2}|a, b \in Q\}$ $Q(\sqrt{2})$, they are infinite sets.

Definition 1.1.1: Fields

We say that a set F together with two binary operations, called addition $+$,$F \times F \to F$, and multiplication $\times$, $F \times F \to F$ satisfying:

- associativity $(a + b) + c = a + (b + c)$
- commutativity $a + b = b + a, ab = ba$
- existence of identities
  there is a $O \in F$ such that $a + O = a$ for all $a \in F$
  there is a $1 \in F$ such that $a \times 1 = a$ for all $a \in F$
- existence of inverses
  for every $a \in F$ there is a $b \in F$ such that $a + b = O$
  for every $a \in F$ there is a $b \in F$ such that $a \times b = 1$
- distributivity
  $a \times (b + c) = a \times b + a \times c$

Example 1: Fields

- For every prime number p, and $n \geq 0$,$n \in N$, there is a field $(F_{p^n}, +, \times)$ with $p^n = q$ elements (cardinality q).
- (C,+,x) is a field

Question: $(R^d, +)$,$d \geq 2$, is there a multiplication, $\times$ such that $(R^d, +, x)$ is a field?

- R^2 , sclar product, $x \cdot y = x_1y_1 + x_2y_2 \in R$ is not a field because $R^2 \times R^2 \to R$.
- cross product, not satisfying Definition 1 and 5.
- d=2, there is a good defination of multiplication, $(R^2, +, \times) \cong (C, +, \times)$.

Lemma 1.1.1

Consider the vector space $(R^2, +)$
Define the multiplication: $(a, b) \times (c, d) = (ac - bd, ad + bc)$
Then $(R^2, +, \times)$ is a field.

Proof:

- You can check that $\times$ is commutative and satisfies associativity and distributivity.
- We can check $(1, 0)$ is a multiplication identity.
- also if $(a, b) \neq (0, 0)$, then

$$(a, b) \times (\frac{a}{a^2 + b^2}, \frac{-b}{a^2 + b^2}) = (\frac{a^2 + b^2}{a^2 + b^2}, \frac{-ab + ab}{a^2 + b^2}) = (1, 0)$$

§1.2 Lecture 2 (02-05)Complex numbers

§1.2.1 The algebra of complex numbers

It is important to classify sets:
in Set Theory: bijection
in Topology: homeomorphism
For **Fields**: isomorphism

Definition 1.2.1: isomorphism

We say that fields F and F' are isomorphic if there exists a funtion

$$i : F \to F'$$

which is a bijection and preserve the binary operations

$$i(a + b) = i(a) + i(b)$$

and

$$i(ab) = i(a)i(b)$$

.
i is called an isomorphism and we write $F \cong F'$

Comments:
There is a unque field up to isomorphism of order (its cardinality)

$$p^n : F_{p^n}$$

Definition 1.2.2: subfield

Given a field (F,+,x), we say that (E,+,x) is a subfield of (F,+,x) if E $\subset$ F and the addition of multiplication of E is the same addition and multiplication of F.

Example 1: subfield

$$Q \subset R$$
$$Q \subset Q(\sqrt{2}) \subset R \subset R^2$$

Definition 1.2.3: complex number

Let F be a field such that:

- R is a subfield of F
- there is an element $i \in F$ which is the root of the equation
$$x^2 + 1 = 0(i^2 = -1)$$

Then we define the field of complex numbers C as

$$C = \{x \in F : x = \alpha + i\beta, \alpha, \beta \in R\}$$

Comments:

- We need to show that there is at least one field F satisfying these properties. **Actually,** $(F, +, \times) = (R, +, \times)$ **satisfies these properties.**
- $R' = \{x \in R : x = (\alpha, 0), \alpha \in R\} \cong R \rightarrow$ R is a subfield of R^2 .
- Define $i := (0, 1) \in R^2$
$$(a, b) \times (c, d) := (ac - bd, ad + bc)$$
Then,
$$i^2 = (0, 1)^2 = (0, 1) \times (0, 1) = (-1, 0)$$

Example 2: exercise

Set of 2×2 real matrices of the form

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix}, a, b \in R$$

This set together with the standard addition and multiplication of matrices is a field and it is isomorphic to $(R^2, +, \times)$.
$F = R^2, C = R^2$

Comments:
It turns out that any defination of C is unique (up to isomorphisms).

Lemma 1.2.1

Let F and F' be two fields satisfying the properties of the definition of complex numbers, then call $\mathbb{C} and \mathbb{C}'$ the corresponding complex number fields.
Then $\mathbb{C} \cong \mathbb{C}'$.

Proof:
Call $i \in \mathbb{C}$ and $i' \in \mathbb{C}'$the square roots -1.
Then:

$$f : \mathbb{C} \to \mathbb{C}' \text{ defined as } f(\alpha + i\beta) = \alpha + i'\beta \text{ is an isomorphism}$$

$$z = \alpha + i\beta, z' = \alpha' + i'\beta', w = \gamma + i\delta, w' = \gamma' + i'\delta'$$

Lemma 1.2.2

Consider the vetor space $(\mathbb{R}, +)$.Then any multiplication '$\times$' defined in this vector space which transforms it into a field $(\mathbb{R}, +, \times)$,is such that $(\mathbb{R}, +, \times) \cong \mathbb{C}$.

Proof:
Consider the elements

$$(1, 0) \text{ and } j = (0, 1) \in \mathbb{R}^2$$

Then any elemnt of $\mathbb{R}^2$ can be written as$\alpha + j\beta, \alpha, \beta \in \mathbb{R}$
Then

$$j^2 = a + jb, a, b \in \mathbb{R}$$

Then,

$$j^2 - jb = a, j^2 - 2\frac{jb}{2} = a \to j^2 - 2\frac{jb}{2} + \frac{b^2}{4} = a + \frac{b^2}{4}$$

$$(j + \frac{b}{2})^2 = a + \frac{b^2}{4} \tag{1.1}$$

Define:$i = \frac{j - \frac{b}{2}}{\sqrt{a + \frac{b^2}{4}}}$
Claim:

$$a + \frac{b^2}{4} < 0$$

If:

$$a + \frac{b^2}{4} \geq 0$$

Then (1.1) can be written as

$$\to ((j - \frac{b}{2}) + \sqrt{a + \frac{b^2}{4}})((j - \frac{b}{2}) - \sqrt{a + \frac{b^2}{4}}) = 0$$

either

$$j - \frac{b}{2} = \sqrt{a + \frac{b^2}{4}}, j - \frac{b}{2} = -\sqrt{a + \frac{b^2}{4}}$$

§1.3 Complex numbers

The field of rationals $\mathcal{Q}$ is the fractional field of the integral domian $\mathcal{Z}$.
With the metrice

$$d(x, y) = |x - y|$$
$$|x| = max\{x, -x\}, \forall x, y \in \mathcal{R}$$

The fielf $\mathcal{Q}$ is not a complete ,metrice space.
Its completion $\mathbb{R}$, nevertheless is not algebraically closed.
For instance, $f(x) = x^2 + 1$, doesn't split in $\mathbb{R}$.
So we define the spliting field for $f(x)$ over $\mathbb{R}$
i.e. the smallest field extension of $\mathbb{R}$ such that $f(x)$ has roots.

Consider the principle ideal $< f(x) >$ in $\mathbb{R}[x]$
since $f(x)$ is irreducible, then $< f(x) >$ si a maximal ideal.
Then the quotient ring $\mathbb{C} = \frac{\mathbb{R}[x]}{<f(x)>}$ is a field!
which is the spilting field of $f(x)$
The field $\mathbb{C}$ has no further spilting extensions! All polynomials in $\mathbb{C}[x]$ split in $\mathbb{C}$.
So as soon as we forced one polynomial. $f(x) = x^2 + 1$ to have roots, we in fact imposed that all polynomials have roots.

Topology:
The Topology of $\mathbb{C}$ is isomorphic to $\mathbb{R}^2$.
Thus $\mathbb{C}$ is locally compact, but not compact space.
The Alexandroff compactification says that we can add a point at infinity to $\mathbb{C}$ so that $\mathbb{C} \cup \{\infty\}$ compact.

Definition 1.3.1: Alexandroff compactification

- A set $U \supseteq \mathbb{C} \cup \{\infty\}$ is an open neighborhood of $\{\infty\}$ if $U = \mathbb{C} \cup \{\infty\}\ K$ When $K \supseteq \mathbb{C}$.

$\mathbb{C} \cup \{\infty\}$ is called the Riemann sphere.

Definition 1.3.2: Holomorphic Functions

A function $f : \Omega \to \mathbb{C}$ is holomorphic where $\Omega \subseteq \mathbb{C}$ is holomorphic iff:

$$\lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} = f'(z_0) \text{ exists}, \forall z_0 \in \Omega$$
$$f(z) = f(x, y) = u(x, y) + iv(x, y), wherez = x + iy \in \Omega$$
$$u(z) = Re(f(z)), v(z) = Im(f(z))$$

$f : \Omega \to \mathbb{C}$ being holomorphic of course implies that $u, v : \Omega \to \mathbb{R}^2$ differentiable.
If we take $f(z_0) = z_0 = 0$ and we write

$$\frac{\partial}{\partial z} = \frac{1}{2}(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}), \frac{\partial}{\partial \bar{z}} = \frac{1}{2}(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y})$$

Then,

$$\frac{f(x)-f(z_0)}{z-z_0} = \frac{f(z)}{z} = (\frac{\partial f}{\partial z})_{(z_0=0)} + \frac{\bar{z}}{z}(\frac{\partial f}{\partial \bar{z}})_{(z_0=0)} + o(z)$$

For real $z \to 0$, we have $\frac{\bar{z}}{z} = 1$, and for purely imaginary $z \to 0$, we have $\frac{\bar{z}}{z} = -1$. we have to restrict $\frac{\partial f}{\partial z} = 0$

Cauchy-Riemann equations:

$$\begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \end{cases}$$

$$f'(z) = \frac{\partial f}{\partial z} = (\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x})(x,y), \text{where } z = x+iy \in \Omega$$

Nevertheless, f being holomorphic is not equivalent to say f satisfies the Cauchy-Riemann equations.
Indeed we take $f(z) = \frac{z^5}{|z|^4}$ on $\mathbb{C}\backslash\{0\}$ and $f(0) = 0$.
We can observe that $f(z)$ satisfies the Cauchy-Riemann equations on $\mathbb{C}$ but f is not even coutinuous at $z = 0$.

Theorem 1.3.1: Looman Menchoff Theorrem

A continuous function $f : \Omega \cup \mathbb{C} \to \mathbb{C}$ is holomorphic iff f satisfies the Cauchy-Riemann equations.

$$\begin{aligned} &\forall f, g \in H(\Omega) \\ &(fg) = f'g + fg' \\ &(\frac{f}{g}) = \frac{f'g - fg'}{g^2} \\ &(g \circ f)'(z) = g'(f(z)) \cdot f'(z) \end{aligned}$$

Chapter 2

Honors Theory of Probability

§2.1 Lecture 1 (02-03)Introduction to Probability

Office Hour: Wed 12:30-1:30pm,Fri 3:30-4:30pm W910

- Homework: weekly
- Grades: 5% participation, 15% homework, 40% midterm, final group project(Max 4 people a group,presetation 20%, Final report 20% 10 pages)

§2.1.1 Intro

Why do we need modern theory of probability?

Example 1

- Coin flip 7 times, what is P[first outcome and fifth outcome are Head]?
  sol:
$$\Omega = (x1, x2, x3, x4, x5, x6, x7)$$
$$A = (x1, x2, x3, x4, x5, x6, x7), x1 = x5 = H$$
$$P(A) = \frac{|\Omega|}{|A|} = \frac{1}{4}$$
- Stock Price (mathematically,geometric Browmion motion, STochastic process, such that $t \to S_t$ is continuous but nowehere differentiable)
  What is P[$S_T \geq 100$]?
$$\Omega = C[0, T] = \{\text{continuous function}, f[0, T] \to R\}$$
$$A = \{f \in C[0, T] | f(t) \geq 100\}$$

  requires measure theory that defines P[$S_T \geq 100$]
  In modern Prob, Probability Space:(Ω, F, P), Ω is sample space, F is σ-algebra(meaningful subset of σ), P is probability measure($P : F \to [0, 1]$).

Topics covered:

- Probability space, σ-algebra, measure ,Conditional Probability and Independence
- Random variables (measurable functions),distribution
- Expectation (Lebesgue integral),Conditional distribution and expectation, functionos of random variables, Radom-Nikodym Derivative
- Random walks

- generating functions, characteristic functions
- Branching processes
- Convergence of random variables, Law of large numbers, Morte-Carlo Method
- Central Limit Theorem
- Time permitting: Large deviations, Markov Chains

§2.1.2 Probability Space

(Ω, F, P)

Example 2: Coinflip and Stock Price

- Coin flip infinite times,

$$\Omega = \{(x1, x2, \cdots), x_i = H, T\}$$

  Let A be the event of 10^6 consecutive Tails,

$$A = \bigcup_{i=1}^{+\infty} \{x_i = x_{i+1} = \cdots = x_{i+10^6-1} = 0, (x_i \in \Omega)\}$$

  $P[A] = 1$

- Discrete Stock Price model, t=0,1,2,$\cdots$,T
  T is maturing time, time step $\Delta t \ll T$

$$N = \frac{T}{\Delta t}$$

$$\text{price go} \begin{cases} \nearrow \text{by factor} : e^{\sigma\sqrt{\Delta t}} \\ \searrow \text{by factor} : e^{-\sigma\sqrt{\Delta t}} \end{cases}$$

$$\Omega = \{(x_1, x_2, \cdots, x_N), x_i = 0 \text{ or } 1\}$$

  Stock price at time t: $\forall \omega \in \Omega$

$$S_N(\omega) = S_0 e^{\sum_{i=1}^{N} x_i \sigma\sqrt{\Delta t} e^{(N - \sum_{i=1}^{N} x_i)(-\sigma\sqrt{\Delta t})}}$$

$$S : \Omega \to R(\text{Random Variable})$$

  Event: return at T is positive but not more than 10%:

$$\{\omega \in \Omega : S_N(\omega) \in (S_0, 1.1S_0]\}$$

Example 3: Gambling

- Gambling:

– start with {0,1,2,$\cdots$} each time bet an interger amount

– if amount of money =0, stays at 0
wealth process: $\Omega - \{0, 1, 2, \cdots\} \times \{0, 1, 2, \cdots\} \times \cdots$
wealth after time n: (random variable) $X_n : \Omega \to N, (x_1, x_2, ...x_n) \to x_n$
$(X_n)$is a Markov Chain (future only depends on present state, but not the past)

$$\text{Event: } \{\text{State j is reached from state i}\} \tag{2.1}$$
$$= \{\omega \in \Omega : \exists n \in N, X_0(\omega) = i, X_n(\omega) = j\} \tag{2.2}$$
$$= \bigcup_{m=1}^{+\infty} \{\omega \in \Omega : X_o(\omega) = i, X_n(\omega) = j\} \tag{2.3}$$

§2.2 Lecture 2 (02-05) – Algebra

§2.2.1 algebra and σ-algebra

in practice, want F to be closed under $\bigcap, \bigcup, ^c$

Definition 2.2.1

Let $\mathcal{A}$ be a collection of subsets of Ω , $\mathcal{A}$ is an algebra iff:

- $\Omega \in \mathcal{A}$
- if A,B$\in \mathcal{A}$ then A$\cap$B $\in \mathcal{A}$
- if A$\in \mathcal{A}$ then A$^c \in \mathcal{A}$

Remark 1

if A,B$\in \mathcal{A} \to$ A $\cap$ B$\in \mathcal{A}$, because A$\cap$B = (A$^c\cup$B^c)^c

Fact:

① $P(\Omega)$(powerset)={A:A $\subset \Omega$} is an algebra

② smallest algebra/trivial algebra: $\{\emptyset, \Omega\}$

③ Let A_1, A_2 be two algebras of Ω

$$A_1 \cap A_2 = \{B \in \Omega : B \in A_1 \text{ and } B \in A_2\} \text{ is an algebra}$$

$$\text{if}(A_j)_{j \in J}\text{is a family of algebras, then } \bigcap_{j \in J} A_j \text{ is an algebra}$$

④ Let $\mathcal{E}$ be any collection of subsets of Ω

$$a(\mathcal{E}) = \bigcap_{\substack{A \supseteq \mathcal{E} \\ A \text{ is an algebra}}} A$$

$a(\mathcal{E})$ is an algebra by ③
in fact, $a(\mathcal{E})$ is the smallest algebra containing $\mathcal{E}$
It is called the algebra generated by $\mathcal{E}$

⑤ Let $A \subseteq \Omega$,$\mathcal{E} = \{A\}$
Then

$$a(\mathcal{E}) = \{\underbrace{A, A^c, \Omega, \emptyset}_{f}\}$$

Proof:

- $a(\mathcal{E}) \subseteq f$
  notice that f is an algebra since $f \supseteq \mathcal{E}$, therefore $f \supseteq a(\mathcal{E})$ because $a(\mathcal{E})$ is the smallest algebra containing $\mathcal{E}$.
- $f \subseteq a(\mathcal{E})$:

$$A \in a(\mathcal{E}), A^c \in a(\mathcal{E})$$

  beacuse $a(\mathcal{E})$ is an algebra, $\emptyset, \Omega \in a(\mathcal{E})$

⑥ $\pi = \{A_1, A_2, \cdots, A_n\}$, $\Omega = \bigcup_{i=1}^{n} A_i, A_i \cap A_j = \emptyset$ Then:

$$a(\pi) = \{\bigcup_{i \in I} A_i, for I \subset 1, 2, \cdots, n\} = \text{ finite disjoint union of } (A_i)_{i=1}^{n}$$

⑦ Let $\mathcal{A}$ be an algebra of $\mathbb{R}$
Let $X : \Omega \rightarrow$ be a function
Then

$$\underbrace{\{X^{-1}(A), A \in \mathcal{A}\}}_{\{\omega \in \Omega : X(\omega) \in A, \text{ for some } A \in \mathcal{A}\}} \qquad \text{is an algebra of } \Omega$$

Hint:

$$X^{-1}(A \cup B) = (X^{-1}(A)) \cup (X^{-1}(B))$$

⑧ $\Omega = \mathbb{R}$, $\mathcal{E} = \{\text{left open right closed intervals}\} = \begin{cases} (a, b], & -\infty \leq a < b < +\infty \\ (a, +\infty) \end{cases}$

Then:

$$\begin{aligned} a(\mathcal{E}) = & \text{ "finite disjioint union of elements in } \mathcal{E}\text{"} \\ = & \underbrace{\{I_1 \cup \cdots \cup I_k, I_j \in \mathrm{E}, I_j \cup I_j = \emptyset(f)\}}_{f} \end{aligned}$$

Hint:

- $f \subseteq a(\mathcal{E})$ strightforward
- $a(\mathcal{E}) \subseteq f$
  - check f is an algebra
  - since $\mathcal{E} \subset f$, $a(\mathcal{E})$ is the smallest algebra containing $\mathcal{E}$, $a(\mathcal{E}) \subseteq f$
  - $(a, b]^c = \underbrace{(-\infty, a] \cup (b, +\infty)}_{\in f}$

Lemma 2.2.1

In Probability, if $(A_k)_{k=1}^{n}$ are disjoint events, we have $P(\bigcup_{k=1}^{+\infty} A_k) = \sum_{k=1}^{+\infty} P(A_k)$

Definition 2.2.2: $\sigma - algebra$

A $\sigma - algebra$ $(\sigma - field)$ A is an collection of subsets of Ω such that

- $\Omega \in A$
- if $A_1, A_2, \cdots \in A$ then $\bigcup_{i=1}^{+\infty} A_i \in A$
- if $A \in A$, then $A^c \in A$

Remark 2

if $A_1, A_2, \cdots \in A$ then $\bigcap_{i=1}^{+\infty} A_i \in A$
σ -algebra represents the collection of information determined by partial derivatives

Example 1

- coinflips infinity many times

$$\Omega = \{(x_1, x_2, \cdots)x_i = 0, 1\} = (0, 1)^{\infty}$$

After observing first outcome, $f_1 = \{\emptyset, \Omega, A_0, A_1\}$
Where $A_0 = \{(0, x_2, x_3, \cdots), x_i = 0, 1\}, A_1 = \{(1, x_2, x_3, \cdots), x_i = 0, 1\}$

After observing second outcome, $f_2 = \{\emptyset, \Omega, A_0, A_1, A_00, A_01, A_10, A_11\}$
Where $A_{00} = \{(0, 0, x_3, x_4, \cdots), x_i = 0, 1\}, A_{01} = \{(0, 1, x_3, x_4, \cdots), x_i = 0, 1\} \cdots$

After observing first n outcomes,

$$f_n = \{\emptyset, \Omega, (A_j)_{j \in (\sigma 1)^n} \text{and finite disjioint unions}\}$$

and

$$f_1 \subseteq f_2 \subseteq \cdots \subseteq f_n \subseteq \cdots$$

Proposition 2.2.0

① intersection of σ-algebras is a σ-algebra

② Let $\mathcal{E}$ be a collecion of subsets of Ω

$$\sigma(\mathcal{E}) = \bigcap_{\substack{f \supseteq \mathcal{E} \\ f \text{ is an algebra}}} f \text{ is a smallest } \sigma\text{-algebra that contains } \mathcal{E}$$

③ For any collection $\mathcal{E}$, we have $a(\mathcal{E}) \subseteq \sigma(\mathcal{E})$

④ For any collection $\mathcal{E}$, we have $\sigma(a(\mathcal{E})) = \sigma(\mathcal{E})$
Hint:

$$a(\mathcal{E}) \subseteq \sigma(\mathcal{E}) \to \sigma(a(\mathcal{E})) \subseteq \sigma(\mathcal{E})$$
$$\mathcal{E} \subseteq \sigma(a(\mathcal{E})) \to \sigma(\mathcal{E}) \subseteq \sigma(a(\mathcal{E}))$$

§2.3 Recitation 1 (02-07) – Problem Solving

§2.3.1 Basic Set Theory

Definition 2.3.1: $\cup, \cap, ^c$

De Morgan's Law:

$$(\bigcup_{j\in J} A_j)^c = \bigcap_{j\in J} A_j^c; (\bigcap_{j\in J} A_j)^c = \bigcup_{j\in J} A_j^c$$

Definition 2.3.2: $\backslash$

$A \backslash B = A \cap B^c$
Then $A \backslash B = A \cap B^c = A \backslash (A \cap B) = B^c \backslash A^c$

Remark 1

$$\bigcap_{n\geq 1} A_n = A_1 \backslash (\bigcup_{j\geq 2} (A_1 - A_j)) \text{ (ex)}$$

§2.3.2 Limits of Sets

Definition 2.3.3: Limit Sets

Let $(A_n)_{n\geq 1}$ be a sequence of sets, then

$$B_k = \bigcup_{n\geq k} A_n, C_k = \bigcap_{n\geq k} A_n$$

Then B_k is increasing, C_k is decreasing
Define:

$$\limsup_{n\to+\infty} A_n = \lim_{k\to+\infty} B_k = \bigcap_{k\geq 1} \bigcup_{n\geq k} A_n$$
$$\liminf_{n\to+\infty} A_n = \lim_{k\to+\infty} C_k = \bigcup_{k\geq 1} \bigcap_{n\geq k} A_n$$

Definition 2.3.4: liminf, limsup of sequence

$$\limsup an = \inf_{k\geq 1} \sup_{n\geq k} a_n, \liminf an = \sup_{k\geq 1} \inf_{n\geq k} a_n$$

When $\limsup A_n = \liminf A_n$, Then we say $\lim_{n\to+\infty} A_n$ exist and $\lim_{n\to+\infty} A_n = \limsup A_n = \liminf A_n$

In probability,

$$\begin{aligned}
\limsup A_n &= \{A_n \text{ occurs infinitely often}\} \\
&= \{An.i.o\} \\
x \in \limsup A_n &\Leftrightarrow \forall k \in \mathbb{N}, \exists n \geq ksuchthatx \in A_n \\
&\Leftrightarrow A_n.i.o \\
\liminf A_n &= \{A_n \text{ occurs eventually}\} \\
&\Leftrightarrow \exists k \in \mathbb{N}, \forall n \geq k, x \in A_n
\end{aligned}$$

Remark 2

1. if A_n increases, then $\limsup A_n = \lim_{n\to\infty} A_n = \bigcup_{n\geq 1} A_n$
   if A_n decreases, then $\liminf A_n = \lim_{n\to\infty} A_n = \bigcap_{n\geq 1} A_n$
2. $$(\limsup A_n)^c = \liminf A_n, (\liminf A_n)^c = \limsup A_n$$

§2.3.3 Exercise

1. $A_k = \begin{cases} E, \text{if k is odd} \\ F, \text{if k is even} \end{cases}$
   Then $\limsup A_n = E \cup F, \liminf A_n = E \cap F$
2. Let $f_n : \mathbb{R} \to \mathbb{R}$, Let A=$\{x \in \mathbb{R} : \lim_{n\to\infty} f_n(x) = f(x)\}$ Then,
   $$\begin{aligned}
   A^c &= \bigcup_{k=1}^{+\infty} \bigcap_{m=1}^{+\infty} \bigcup_{n=m}^{+\infty} \{x : |f_n(x) - f(x)| \geq \frac{1}{k}\} \\
   &= \bigcup_{k=1}^{+\infty} [\limsup_n \{x : |f_n(x) - f(x)| \geq \frac{1}{k}\}]
   \end{aligned}$$
3. Suppose that $\lim_{n\to\infty} f_n(x) = f(x), \forall x \in \mathbb{R}$
   Then
   $$\begin{aligned}
   \{x : f(x) \leq t\} &= \bigcap_{k=1}^{+\infty} \bigcup_{m=1}^{+\infty} \bigcup_{n=m}^{+\infty} \{x \in \mathbb{R} : f_n(x) < t + \frac{1}{k}\} \\
   &= \bigcap_{k=1}^{+\infty} [\liminf_n \{x : f_n(x) < t + \frac{1}{k}\}]
   \end{aligned}$$

Proof:

1. ex2: Want

$$A = \bigcup_{k\geq 1}^{+\infty} \bigcap_{\geq =1}^{+\infty} \bigcup_{n\geq m}^{+\infty} \{x : |f_n(x) - f(x)| \geq \frac{1}{k}\}$$
$$f_n(x) \to f(x) \text{ iff } \forall \epsilon > 0, \exists N \geq N, |f_n(x) - f(x)| < \epsilon$$
$$\{x : f_n(x) \to f(x)\} = \bigcap_{\epsilon>0} \bigcup_{N} \bigcap_{n\geq N} \{x : |f_n(x) - f(x)| < \epsilon\},$$
$$\text{USE that } \{|f_n - f| < \epsilon\} \text{ is monotone increasing in } \epsilon$$

Review on mapping:

$$f : X \to Y, f^{-1} : Y \to X$$

Basic properties:

- $f(\bigcup_{i\in I}) = \bigcup_{i\in I} f(A_i)$
- $f(\bigcap_{i\in I}) = \bigcap_{i\in I} f(A_i)$

For (B_i) subset of Y:

- if $B_1 \subset B_2, f^{-1}(B_1) \subset f_{-}1(B_2)$
- $f^{-1}(\bigcup_{i\in I} B_i) = \bigcup_{i\in I} f^{-1}(B_i)$
- $f^{-1}(B^c) = (f^{-1}(B))$

Definition 2.3.5: Indicator Mapping

$$1_A : X \to \{0, 1\}$$
$$1_A(x) = \begin{cases} 1, x \in A \\ 0, x \notin A \end{cases}$$

Exercise:

1. $1_{limsupA_n} = limsup1_{A_n}$
2. $1_{liminfA_n} = liminf1_{A_n}$

Proof: if

$$\begin{aligned} 1_{limsupA_n}(x) = 1 &\Leftrightarrow x \in limsupA_n \\ &\Leftrightarrow \forall k \in \mathbb{N}, \exists n \geq k, x \in A_n \\ &\Leftrightarrow \forall k \in \mathbb{N}, \exists n \geq k, 1_{A_n}(x) = 1 \\ &\Leftrightarrow limsup1_{A_n}(x) \geq 1 \text{(Definition of limsup)} \\ &\Leftrightarrow 1_{limsupA_n}(x) = 1 \end{aligned}$$

Exercise: Let $\mathcal{A}_1, \mathcal{A}_2$ be algebras of Ω

1. show that $\underbrace{\mathcal{A}_1 \cap \mathcal{A}_2}_{B\subset\Omega: B\in\mathcal{A}_1, B\in\mathcal{A}_2}$ is an algebra

2. show that $\underbrace{\mathcal{A}_1 \cup \mathcal{A}_2}_{B\subset\Omega: B\in\mathcal{A}_1 \text{ or } B\in\mathcal{A}_2}$ is an algebra iff $\mathcal{A}_1 \subseteq \mathcal{A}_2$ or $\mathcal{A}_2 \subseteq \mathcal{A}_1$

Proof: Suppose by contradiction that

$$\exists A_1 \in \mathcal{A}_1 \text{ but } A_1 \notin \mathcal{A}_2 \text{ and } A_2 \in \mathcal{A}_2 \text{ but } A_2 \notin \mathcal{A}_1 \text{ and } \mathcal{A}_1 \cup \mathcal{A}_2 \text{ is an algebra}$$

Therefore:

$$A_1 \cup A_2 \in \mathcal{A}_1 \cup \mathcal{A}_2, A_1 \backslash A_2 = A_1 \cap \underbrace{A_2^c}_{\mathcal{A}_1\cup\mathcal{A}_2} \in \mathcal{A}_1 \cup \mathcal{A}_2$$

$$A_2 \backslash A_1 = A_2 \cap A_1^c \in \mathcal{A}_1 \cup \mathcal{A}_2$$

$\Rightarrow$ at least two of

$$(A_1 \backslash A_2) \cup (A_2 \backslash A_1), A_1 \backslash A_2, A_2 \backslash A_1 \text{ are in } \mathcal{A}_1 \text{ or } \mathcal{A}_2$$

Assume in $\mathcal{A}_1$

$\Rightarrow$ (by $\mathcal{A}_1$ is an algebra, all three sets are in $\mathcal{A}_1$)

$$\Rightarrow A_2 = (\underbrace{A_1 \cup A_2}_{\in\mathcal{A}_1}) \backslash (\underbrace{A_1 \backslash A_2}_{\mathcal{A}_1}) \in \mathcal{A}_1$$

Contradiction!

Chapter 3

Partial Differential Equations

§3.1 Lecture 1 (02-03)Initial and Boundary Conditions

§3.1.1 What are the PDEs?

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} - u \tag{3.1}$$

$$u = u(x, y, t) \tag{3.2}$$

(3.1) is a second order PDE

§3.1.2 Classical solutions

To simplify, we consider here equations of one unknown function u of one or more variable

By a solution we mean a sufficiently smooth function u that satisfies the PDE at every point of the domain of its definition(D).

Sometimes, we will have to impose some boundary conditions on the domain D.

$C^n = \{u : D \to R :$ n-times continuous differentiable w.r.t all variables $\}$ n is the order of the equation

Example 1

- Heat equation:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$
$$u = u(x, t)$$
$$(t, x) \in D \subset R^2$$
$$u(t, x), \frac{\partial u}{\partial t}, \frac{\partial u}{\partial x}, \frac{\partial^2 u}{\partial x^2}, \frac{\partial^2 u}{\partial t^2}$$
$$n = 2$$
$$\frac{\partial^2 u}{\partial x \partial t} = \frac{\partial^2 u}{\partial t \partial x} \in C^2$$
$$u \in C^2(D)$$

- Some Solution:

$$u(t, x) = t + \frac{x^2}{2}, D = R^2$$

$$u(t,x) = \frac{e^{-\frac{x^2}{4t}}}{2\sqrt{\pi t}}, D = \{(t,x) : t > 0\}$$
$$u(t,x) = e^{-t+ix} = e^{-t}cos(x) + ie^{-t}sin(x), D = R^2$$

§3.1.3 Initial and Boundary Conditions

What is Initial data?
$u(0,x) = u_0(x)$

What are Boundary Conditions? rigurous definition of Boundary

- Dirichlet Boundary Condition: $u(t,x)|_{\partial D} = g(t,x)$
- Neumann Boundary Condition: When the normal derivative of u is specified $\frac{\partial u}{\partial n}$ where n is the normal to the boundary $\partial(D)$
$$\frac{\partial u}{\partial n}(t,x) = g(t,x)$$
- Mixed Boundary Condition: Combination of Dirichlet and Neumann

§3.1.4 Linear and Nonlinear waves

Stationary waves

$$\frac{\partial u}{\partial t} = 0 \tag{3.3}$$
$$u = u(t,x) \tag{3.4}$$

This is a first order linear homogeneuous PDE
Solution of the form $u = u(t) = c, \forall c \in R$
Let's integral (3.3) :
$$0 = \int_0^t \frac{\partial u}{\partial t}(s,x)ds = u(t,x) - u(0,x)$$

Thus if an initial conditional
$$u(0,x) = f(x)$$
$$u(t,x) = f(x), \forall t, x$$

Thus for any given $f \in C^1(R)$, we have a solution of (3.3) and this is a stationary wave
The domain of this solution is R if f is defined everywhere on R
Power of domain to change the solution:
Consider the ODE:
$$\frac{\mathrm{d}u}{\mathrm{d}t} = 0, D = (-\infty, 0) \cup (0, \infty)$$

The solution is:
$$u(t) = \begin{cases} c_1, t < 0 \\ c_2, t > 0 \end{cases} \quad \forall c_1, c_2 \in R$$

similarly, for the PDE in example (3.3):
$$u(t,x) = \begin{cases} 0, x > 0 \\ x^2, x \leq 0, t > 0 \\ -x^2, x \leq 0, t < 0 \end{cases}$$

it is a $\mathbb{C}^1$ solution of (3.3) on $\mathbb{R}^2$ and it is a classical solution.

§3.2 Lecture 1 (02-05)Transport and Traveling waves

§3.2.1 uniform transport

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0 \tag{3.5}$$

$$u = u(0,x) = f(x) \tag{3.6}$$

$$x \in \mathbb{R}, t_0 = 0, t - t_0 = t, c \in \mathbb{R}$$

use change of variable:

$$\xi = x - ct$$

ξ is called characteristic variable
let's look for a solution of (3.5) under the form

$$u(t,x) = v(t,\xi) = v(t, x - ct)$$

use chain rule:

$$\frac{\partial u}{\partial t} = \frac{\partial v}{\partial t} - c\frac{\partial v}{\partial \xi}$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial \xi}$$

Thus:

$$0 = \frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = \frac{\partial v}{\partial t} - c\frac{\partial v}{\partial \xi} + c\frac{\partial v}{\partial \xi} = \frac{\partial v}{\partial t}$$

Conclusion:

$$\frac{\partial v}{\partial t} = 0$$

Solution is constant $v(\xi)$

$$u(t,x) = v(\xi) = v(x - ct)$$

Proposition 3.2.0

if u solves (3.5), then u has the form $u(t,x) = v(x - ct)$ where v is a function of one variable that is $\mathbb{C}^1$-smooth.
Acutually, v is satisfying

$$u(0,x) = v(x) = f(x)$$

Conclusion:

$$\begin{cases} \frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0 \\ u = u(0,x) = f(x) \end{cases}$$

The solution is

$$u(t,x) = f(x - ct)$$

$f(x - ct)$ is a traveling wave solution moving at velocity c:
if c>0, the wave moves to the right
if c<0, the wave moves to the left
if c=0, the wave is stationary

Example 1

$$\begin{cases} \frac{\partial u}{\partial t} + 2\frac{\partial u}{\partial x} = 0 \\ u = u(0,x) = \frac{1}{1+x^2} \end{cases}$$

The solution is

$$u(t,x) = \frac{1}{1+(x-2t)^2}$$

The lines $x = ct + k$ are called the characteristic lines for the equation (3.5)
on the characteristic lines, the solution is always constant
on geometric interpretation, $u(t,x) = f(x-ct) = f(k)$

Consider the equation:

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} + au = 0 \tag{3.7}$$

where a and c are given numbers.
Use change of variable:

$$u(t,x) = v(t,\xi) = v(t, x-ct)$$

we will get by the same computation

$$\frac{\partial v}{\partial t} + av = 0$$

The solution will be

$$v(\xi) = e^{-at}f(\xi)$$
$$u(t,x) = v(t,\xi) = v(x-ct) = e^{-at}f(x-ct)$$

So,

$$\begin{cases} u(t,x) = e^{-at}f(x-ct) \\ u(0,x) = f(x) \end{cases}$$

§3.2.2 Nonuniform transport

$$\frac{\partial u}{\partial t} + c(x)\frac{\partial u}{\partial x} = 0 \tag{3.8}$$

In the uniform case(c constant), characteristic line is x=ct+k (the solution of ODE below)

$$\frac{\mathrm{d}x}{\mathrm{d}t} = c$$

In the general non-uniform case, we can still define characteristic curves

$$\frac{d}{dt}x(t) = c(x(t))$$

Let's show that the solutions of (3.8) are constants along the characteristic curves (t,x(t))

We need to check that $u(t, x(t)) = h(t)$ is constant of t
compute the derivative of $h(t)$

$$\begin{aligned}\frac{d}{dt}h(t) &= \frac{d}{dt}u(t,x(t)) = \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x}\cdot\frac{\mathrm{d}x}{\mathrm{d}t}\\ &= \frac{\partial u}{\partial t} + c(x(t))\cdot\frac{\partial u}{\partial x} = 0\end{aligned}$$

h(t) and thus u(t,x(t)) is constant along the characteristic curves

Definition 3.2.1

The graph of a solutioni of the ODE

$$\frac{d}{dt}x(t) = c(x(t))$$

is called a characteristic curve for the transport equation (3.8) with speed c(x)

Thus at each point (t,x) the slope of the characteristic curve is c(x) (the wave speed)
In the particular case c(x)=c, the characteristic curves are straight lines of slope c
Moreover,
(3.8) is an autonomous first order ODE which can be "solved" using the method of separation of variables
Assume that c(x)$\neq$0

$$\begin{aligned}\frac{dx(t)}{dt} &= c(x(t))\\ \frac{dx}{c(X)} &= dt \rightarrow\\ \beta(x) = \int \frac{dx}{c(x)} &= t + k\end{aligned}$$

Thus characteristic curves satisfies the implicit equatioin

$$\beta(x) = t + k$$

We can find x from the equality using β^{-1}(the inverse of β)

$$\begin{aligned}\beta(x) &= t + k\\ \beta^{-1}\beta(y) &= y\\ x(t) &= \beta^{-1}(t+k)\end{aligned}$$

Remark 1

How to determine where $c(x_0) = 0$?
Solve this ODE

$$\begin{cases}\frac{dx}{dt} = c(x)\\ x(t_0) = x_0\end{cases}$$

and $x(t) = x_0$ wil be the characteristic curve going through (t_0, x_0)

From the proposition, we know that $u(t, x(t))$=constant

$$\begin{aligned} u(t, x(t)) &= v(k) \\ &= v(\beta(x(t)) - t) \end{aligned}$$

where k is the constant determining the characteristic curve (t,x(t))

$$\begin{gathered} u(t, x) = v(\xi) \\ \text{where } \xi = \beta(x) - t \end{gathered}$$

which we will call characteristic variable

$$u(t, x) = v(\beta(x) - t) \tag{3.9}$$

This formula way fail at points where c vanishes

$$\begin{aligned} t &= 0 \\ u(0, x) = f(x) &= v(\beta(x) - 0) \\ f(x) &= v(\beta(x)) \\ v(\xi) &= f(\beta^{-1}(\xi)) \\ &= f \cdot \beta^{-1}(\xi) \\ u(t, x) &= f \cdot \beta^{-1}(\beta(x) - t) \end{aligned}$$

This is the formula for the solution of (3.8) in the general case
which is:

$$\begin{cases} \frac{\partial u}{\partial t} + c(x)\frac{\partial u}{\partial x} = 0 \\ u(0, x) = f(x) \end{cases}$$

Chapter 4

Algorithm

§4.1 Lecture 1 (02-03)Asymptotic bounds and big O

Definition 4.1.1: big-Theta notation(=)

Let $f, g : N \to R^+$, we say that $f(n) = \Theta(g(n))$ if there $\exists\ a, b > 0$ and $n_0 \in \mathbb{N}$such that $a \cdot g(n) \leq f(n) \leq b \cdot g(n)$ for all $n > n_0$.

Intuitively, $f(n) = \Theta(g(n))$ means that $f(n)$ grows as fast as $g(n)$ up to a constant when n tends to infinity.

Examples: $3n = \Theta(n)$, $2n^2 + 7n = \Theta(n^2)$, $\ln n^2 = \Theta(\log n)$

Definition 4.1.2: big-O notation(≦)

Let $f, g : N \to R^+$, we say that $f(n) = O(g(n))$ if $\exists\ n_0 \in \mathbb{N}$ and $b > 0$ such that $f(n) \leq b \cdot g(n)$ for all $n > n_0$.
$\lim_{n\to\infty} sup \frac{f(n)}{g(n)} < +\infty$

Definition 4.1.3: big-Omega notation(≥)

Let $f, g : N \to R^+$, we say that $f(n) = \Omega(g(n))$ if $\exists\ n_0 \in \mathbb{N}$ and $a > 0$ such that $a \cdot g(n) \leq f(n)$ for all $n > n_0$.
$\lim_{n\to\infty} inf \frac{f(n)}{g(n)} > 0$

Remark 1

$$f(n) = \Theta(g(n)) \Leftrightarrow f(n) = O(g(n)) \text{ and } f(n) = \Omega(g(n))$$
$$f(n) = O(g(n)) \Leftrightarrow g(n) = \Omega(f(n))$$

Definition 4.1.4: ittle-o notation(<)

We write $f(n) = o(g(n))$ if for any $b > 0$ there $\exists\ n_0 \in \mathbb{N}$ such that $f(n) < b \cdot g(n)$ for all $n > n_0$.

Definition 4.1.5: little-omega notation(>)

We write $f(n) = \omega(g(n))$ if for any $a > 0$ there $\exists n_0 \in \mathbb{N}$ such that $a \cdot g(n) < f(n)$ for all $n > n_0$.

Remark 2

$f(n) = o(g(n)) \Rightarrow f(n) = O(g(n))$ and $f(n) \neq \Theta(g(n))$
$f(n) = \omega(g(n)) \Rightarrow f(n) = \Omega(g(n))$ and $f(n) \neq \Theta(g(n))$
$f(n) = \omega(g(n)) \Leftrightarrow g(n) = \omega(f(n))$

$$n^{1+e}, nlog(n), e \to 0$$

Theorem 4.1.1

Any Algorithm has to check at least log(n+1) entries of the sorted array for searching an element in the worst case

§4.2 Recitation 1 (02-07)–Problem Solving

Notation	Bounding Conditions	Relation	Type
$T(n) = O(f(n))$	$\exists c > 0, \exists n_0, \forall n \geq n_0, T(n) \leq cf(n)$	$\leq$	O
$T(n) = \Omega(f(n))$	$\exists c > 0, \exists n_0, \forall n \geq n_0, T(n) \geq cf(n)$	$\geq$	Ω
$T(n) = \Theta(f(n))$	$\exists c_1, c_2 > 0, \exists n_0, \forall n \geq n_0, c_1 f(n) \leq T(n) \leq c_2 f(n)$	$=$	Θ
$T(n) = o(f(n))$	$\forall c > 0, \exists n_0, \forall n \geq n_0, T(n) < cf(n)$	$<$	o
$T(n) = \omega(f(n))$	$\forall c > 0, \exists n_0, \forall n \geq n_0, T(n) > cf(n)$	$>$	ω

Possible useful formula:

$$\int_1^n x^k dx = \frac{n^{k+1}}{k+1} - \frac{1}{k+1}$$

1. Multiplication Rule. Prove that for $h(n) = \Omega(1)$, if $f(n) = \Theta(g(n))$ then $f(n) \cdot h(n) = \Theta(g(n) \cdot h(n))$ (The same also holds for $O, \Omega, \omega, \omega$)
   Solution:

   Since $f(n) = \Theta(g(n)), \exists c_1, c_2 > 0, \exists n_0$ such that
   $c_1 g(n) \leq f(n) \leq c_2 g(n), \forall n \geq n_0$
   Since $h(n) = \Omega(1),$ there exist constants $c_3 > 0$ and n_1 such that
   $h(n) \geq c_3, \forall n \geq n_1$
   Combining the two, we have:
   $c_1 g(n)h(n) \leq f(n)h(n) \leq c_2 g(n)h(n)$
   Therefore $f(n)h(n) = \Theta(g(n)h(n))$

2. Addition Rule.Let $f_1, ..., f_k : N \to R_{\geq 0}$ for a fixed number k and $f(n) = \sum_{i=1}^{k} f_i(n)$ .Prove that if $f_i(n) = O(f_1(n))$ for all $i \in \{1, \ldots, k\}$,then $f(n) = \Theta(f_1(n))$ .Here

f_1 can be replaced with any one of $f_1, ..., f_k$
Solution:

Since $f_i(n) = O(f_1(n)),$ there exist constants c_i and n_0 such that
$f_i(n) \leq c_i f_1(n), \forall n \geq n_0$
Therefore,
$\sum_{i=1}^{k} f_i(n) \leq \sum_{i=1}^{k} c_i f_1(n) = Cf_1(n)$ where $C = \sum_{i=1}^{k} c_i$

Additionally, there exists at least one $f_j(n) = \Omega(f_1(n)),$ so $\sum_{i=1}^{k} f_i(n) = \Theta(f_1(n))$

3. Let f and g be non-negative functions. Define $T(n) = max(f(n), g(n))$ .Prove that $T(n) = \Theta(f(n) + g(n))$
Solution:

   Since $max(f(n), g(n)) \leq f(n) + g(n),$ we have $T(n) = O(f(n) + g(n))$
   Also, $f(n) + g(n) \leq 2max(f(n), g(n)),$ so $T(n) = \Omega(f(n) + g(n))$
   Therefore, $T(n) = \Theta(f(n) + g(n))$

4. Let f and g be non-decreasing real-valued functions defined on positive integers and $f(1) \geq 1$ and $g(1) \geq 1$ .Prove or disprove that $f(n) = O(g(n)) \Rightarrow 2^{f(n)} = O(2^{g(n)})$
Solution:

   Since $f(n) = O(g(n)),$ there exists a constant $c > 0$ and n_0 such that
   $f(n) \leq c \cdot g(n), \forall n \geq n_0$
   Therefore,
   $2^{f(n)} \leq 2^{c \cdot g(n)}$
   Since 2^c is constant, we conclude that $2^{f(n)} = O(2^{g(n)})$

5. Prove that $log(n!) = \Theta(nlogn)$ .Hint:To show an upper bound,compare $n!$ with n^n .To show a lower bound, compare it with $\left(\frac{n}{2}\right)^{\frac{n}{2}}$
Solution:

   We know that:
   $log(n!) = log(1) + log(2) + \ldots + log(n)$
   Since $log(n)$ is increasing, the sum can be bounded by:
   $log(n!) \leq \sum_{k=1}^{n} log(n) = nlog(n)$
   From the bound, we can use the fact that $log(k) \geq log(\frac{n}{2})$ for all $k \geq 2$
   Hence, we can sum the lower half of the terms to get:
   $log(n!) \geq \sum_{k=\frac{n}{2}}^{n} log(k) \geq \sum_{k=\frac{n}{2}}^{n} log(\frac{n}{2}) = \frac{n}{2}log(\frac{n}{2})$

6. Show that $c^n = o(n!)$ for any constant C
   Solution:

   We need to prove that
   $$\lim_{n\to\infty} \frac{c^n}{n!} = 0$$
   Using the recursive relation:
   $$\frac{c^n}{n!} = \frac{c^{n-1}}{(n-1)!} \times \frac{c}{n}$$
   As $n \to \infty, \frac{c}{n} \to 0,$ Thus,
   $$\lim_{n\to\infty} \frac{c^n}{n!} = 0$$

7. Show that $c^{n^{1.001}} = \omega(n!)$ for any constant C .Hint: For any fixed numbers $p, q > 0$,we have $n^p = \omega(log^q n)$
   Solution:

   We need to prove that
   $$\lim_{n\to\infty} \frac{c^{n^{1.001}}}{n!} = \infty$$
   Taking the algorithm, we have:
   $$log(\frac{c^{n^{1.001}}}{n!}) = n^{1.001} log(c) - log(n!)$$
   Since $log(n!) \approxeq nlogn,$ and $n^1.001$ grows faster than $nlogn,$ we conclude that
   $$\lim_{n\to\infty} log(\frac{c^{n^{1.001}}}{n!}) = \infty$$
   So, $\lim_{n\to\infty} \frac{c^{n^{1.001}}}{n!} = \infty$
   Therefore, $c^{n^{1.001}} = \omega(n!)$

8. Show $\sum_{i=1}^{n} i^k = \Theta(n^{k+1})$ .Hint: Show that $n^{k+1} - (n-1)^{k+1} = \Theta(n^k)$
   Solution:

   Upperbound: Since $i^k \leq n^k$, we have
   $$\sum_{i=1}^{n} i^k \leq \sum_{i=1}^{n} n^k = n^{k+1}$$
   Lowerbound: Using the integral approximation
   $$\sum_{i=1}^{n} i^k \geq \int_1^n x^k dx = \frac{n^{k+1}}{k+1} - \frac{1}{k+1}$$
   Therefore,
   $$\sum_{i=1}^{n} i^k = \Theta(n^{k+1})$$

9. Search for the peak. Suppose we have an array $A[1...n]$ and we know that there exists some k (value unknown) such that $A[1] < A[2] < ... < A[k]$ and $A[k] > A[k+1] > \ldots > A[n]$ .We call $A[k]$ the peak of A .Design an algorithm to find the peak and analyze it in $O(logn)$ time.
   Solution:

   Use Binary Search Algorithm:
   1.Set $l = 1, r = n$
   2.While $l < r$ do
   3.Set $m = \left\lfloor \frac{l+r}{2} \right\rfloor$
   4.If $A[m] < A[m+1]$ then $l = m+1$
   5.Else $r = m$
   6.Return l
   Analysis:
   Let $T(n)$ be the time complexity of the algorithm
   Let $n_0 = 1$
   Let $c = 1$
   Let $T(n) = T\left(\frac{n}{2}\right) + c$
   Let $T(n) = O(logn)$

10. 3SUM in a sorted array.Let $A[1 \ldots n]$ be a sorted array. We want to see $A[i] + A[j] + A[k] = x$ for some different i, j, k ,where x is a given number. Design an algorithm $3SUM(n, a, x)$ which can return (i, j, k) or 0 if no such three elements exist.How fast can this problem be solved?

Solution:

fix one element A[i], then use two pointers on the remaining array to find $A[j] + A[k] = x - A[i]$
The time complexity is $O(n^2)$

Algorithm 3SUM(n,a,x):

```
Input: Array A of n intergers, integer x
Output: indices i,j,k such that A[i]+A[j]+A[k]=x or return "No solution" if no such triplet exists
for i=0 to n-1 do
 //Avoid duplicates by skipping repeated elements
 if i>0 and A[i]==A[i-1]
 continue
 left=i+1,right=n-1
 while left<right do
 sum=A[i]+A[left]+A[right]
 if sum==x
 return i,left,right //Found the triplet
 if sum<x
 left=left+1 //increase the sum by moving left pointer to the right
 else
 right=right-1 //decrease the sum by moving right pointer to the left
return "No solution" //no such triplet exists
```